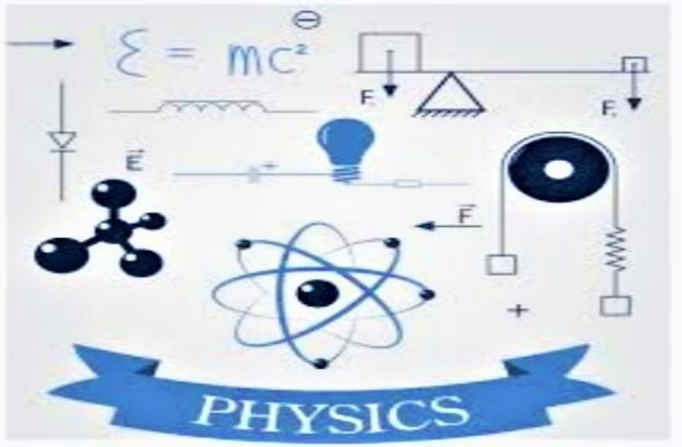
Fizika

9-sinf

Mavzu:

Masalalar yechish.



Yoqilg‘ining solishtirma yonish issiqligi.

$$Q = qm$$

$$q = \frac{Q}{m} \Rightarrow 1\ \frac{J}{kg}$$

Termodinamikaning birinchi qonuni

$$Q = \Delta U + A$$

Izobarik jarayonda: $Q = \Delta U + A$

Izoxorik jarayonda: $V = const.\ Q = \Delta U$

Izotermik jarayonda: $T = const.\ Q = A$

Adiabatik jarayonda: $Q = 0,\ A = -\Delta U$

17-test 76-bet

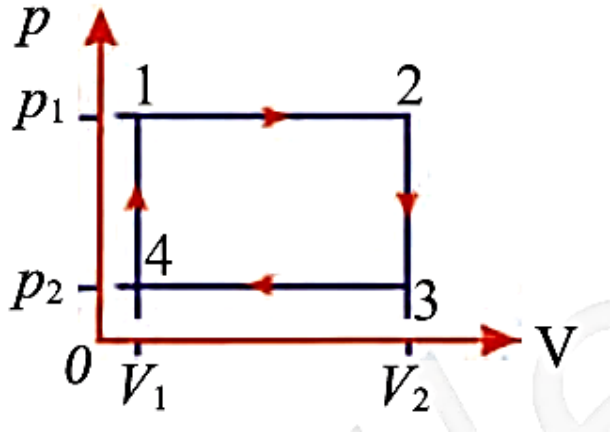
Ideal gazning rasmda ko‘rsatilgan siklni o‘tishda bajargan ishini hisoblab toping.

A) $(p_1 - p_2)(V_2 - V_1)$

B) $p_1(V_2 - V_1)$

C) $p_2(V_2 - V_1)$

D) $(p_2 - p_1)V_2$



To‘g‘ri javob: A) $(p_1 - p_2)(V_2 - V_1)$

17-mashq 1-masala

Izotermik jarayonda gazga **5 kJ** issiqlik berilgan bo‘lsa, gaz ustida qancha ish bajarilgan bo‘ladi?

Berilgan:

$T = const.$

$Q = 5\ \text{kJ}$

$A - ?$

Formula:

$$Q = \Delta U + A$$

$T = const.$ da $\Delta T = 0$

$$\Delta U = 0$$

$$A = Q$$

Yechish: $A = 5\ \text{kJ}$

Javob: $A = 5\ \text{kJ}$

17-mashq 4-masala

Normal sharoitda bir atomli ideal gazga issiqlik berilganda gaz izobarik ravishda $0{,}05\,m^3$ ga kengayadi. Gazning ichki energiyasi qanday o‘zgaradi?

Berilgan:

$p = const$

$p = 10^5 Pa$

$\Delta V = 0{,}05\ m^3$

$\Delta U - ?$

Formula:

$$Q = \Delta U + A$$

$$\Delta U = Q - A \qquad A = p\Delta V$$

bir atomli gaz uchun $p = const$. da

$$Q = \frac{5}{2}A \Rightarrow \Delta U = \frac{5}{2}A - A = \frac{3}{2}A$$

$$\Delta U = \frac{3}{2}p\Delta V$$

Yechish: $\Delta U = \frac{3}{2} \cdot 10^5 Pa \cdot 0{,}05\ m^3 = 7500$ J$= 7{,}5$ kJ

Javob: $\Delta U = 7,5$ **kJ**

17-mashq 5-masala

Metall ballondagi 25 mol bir atomli ideal gazning temperaturasini 20 K ga orttirish uchun unga qancha issiqlik berish kerak?

Berilgan:

$V = const.$

$\nu = 25\ mol$

$\Delta T = 20\ \text{K}$

$R = 8{,}3\ \frac{J}{K \cdot mol}$

$Q - ?$

Formula:

$$Q = \Delta U + A$$

$V = const.$ da $Q = \Delta U$

bir atomli gaz uchun

$$\Delta U = \frac{3}{2}\nu R \Delta T \ \Rightarrow \ Q = \frac{3}{2}\nu R \Delta T$$

Yechish: $\text{Q} = \frac{3}{2} \cdot 25\ mol \cdot 20\ K \cdot 8{,}3\ \frac{J}{K \cdot mol} = 6225\ \text{J}$

Javob: Q = 6225 J

17-mashq 6-masala

Silindrik idishdagi erkin siljiydigan porshen ostida bir atomli gaz bor. Gaz bosimi $10^5\ Pa$.

Gazga qancha issiqlik berilsa, gazning hajmi $2\ l$ ga oshadi?

Berilgan:

$p = const.$

$p = 10^5\ Pa$

$\Delta V = 2\ l = 2 \cdot 10^{-3}\ m^3$

Topish kerak: $Q - ?$

Formula:

$$Q = \Delta U + A$$

bir atomli gaz uchun $p = const.$ da

$$Q = \frac{5}{2}A \qquad A = p\Delta V$$

$$Q = \frac{5}{2}p\Delta V$$

Yechish: $\mathrm{Q} = \frac{5}{2} \cdot 10^5\ Pa \cdot 2 \cdot 10^{-3}\ m^3 = 500\ \mathrm{J}$

Javob: Q = 500 J

Mustaqil bajarish uchun topshiriqlar

1. 17-mashq 2, 3, 7-masalalarni yechish. (71-bet)
2. 77-bet 21-dan 29-gacha bo‘lgan testlarni yechish.